

Supplementary Appendix to:

RP1 Combined With Nivolumab in Advanced Anti-PD-1–Failed Melanoma (IGNYTE)

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Contents

SUPPLEMENTAL METHODS	3
Additional Courses of RP1	3
Definitions of Injected Tumors.....	3
Immunohistochemistry Analysis	4
RNA Sequencing Analysis	4
Injected vs Non-Injected Lesion Analysis	6
SUPPLEMENTAL FIGURES AND TABLES	7
FIG S1. IGNYTE study design.	7
FIG S2. CONSORT diagram.	8
FIG S3. Patient examples of the outcome of RP1 plus nivolumab.	9
FIG S4. RP1 combined with nivolumab increases CD8+ T-cell infiltration and PD-L1 expression.....	13
TABLE S1. Summary of Dose Volume in the Initial Course.....	16
TABLE S2. Summary of Best Reduction in Injected and Non-Injected Lesions Among Responding Patients by RECIST 1.1.	17
TABLE S3. Summary of Tumor Reduction in Non-Injected Lesions in Visceral Organs of Responding Patients by RECIST 1.1.	18
TABLE S4. Summary of Responses by Injection Type by RECIST 1.1..	19
TABLE S5. BOR Among All Patients and Across Subgroups by Blinded Independent Central Review According to mRECIST 1.1.	20
TABLE S6. Full List of Grade 3/4 TRAEs.	22
REFERENCES.....	24

SUPPLEMENTAL METHODS

Additional Courses of RP1

Additional courses of RP1 (vusolimogene oderparepvec; up to eight doses per course) could be administered if the following criteria were met: the patient had injectable disease and lesions that were safe to inject, the patient had a stable performance status and was deriving clinical benefit in the opinion of the investigator, and the patient did not experience any of the following: a grade 4 adverse event (AE), a grade 3 injection-site reaction lasting >14 days, a grade 3 suspected immune-mediated AE, or a grade 3 infection lasting >14 days (unless deemed unrelated to RP1) during the initial course of RP1. For patients who received additional RP1, additional doses were administered at 1×10^7 plaque-forming units (PFU)/mL every 2 weeks either in combination with nivolumab or as monotherapy if nivolumab was stopped due to nivolumab-related toxicity.

Definitions of Injected Tumors

Superficial tumors were defined as tumors that could be visualized or palpated and accessed with standard-size needles and syringes. Guidance by ultrasound could be used if necessary. Deep/visceral tumors were defined as tumors that could not be directly observed or palpated and required imaging guidance (eg, ultrasound, computed tomography, endoscopy, bronchoscopy) to correctly locate and inject the tumors.

Visceral tumors are deep tumors associated with visceral organs.

Immunohistochemistry Analysis

Tumor biopsies were collected at screening or cycle 1 day 1 (either archival within 90 days or fresh tissue prior to dosing) and at day 43. Formalin-fixed paraffin-embedded material was used for staining with routine hematoxylin and eosin (Richard-Allan Scientific, USA #7211/7111), CD8 immunohistochemistry (SP57, Roche Tissue Diagnostics, USA), and programmed death-ligand 1 (PD-L1) immunohistochemistry (28-8 pharmDx, Dako North America, Inc., USA). Glass coverslips were applied to the stained slides. Slides were reviewed by a board-certified pathologist for the following criteria: (1) overall presence/absence of tumor, (2) PD-L1 tumor proportion score and combined positive score, (3) PD-L1 intra- and peri-tumoral staining using a semi-quantitative scale, and (4) CD8 intra- and peri-tumoral staining using a semi-quantitative scale. Immunohistochemistry analysis was performed at DCL Pathology (USA).

RNA Sequencing Analysis

The concentration and integrity of the extracted total RNA were estimated using the Quant-iT™ RiboGreen® RNA Assay Kit (Thermo Fisher Scientific, USA) and the Fragment Analyzer System using the RNA High Sense chip (Agilent, USA), respectively. Following a quality control check of the RNA, up to 30 ng of RNA with DV₂₀₀ >20% was taken into a standard library preparation protocol using SMARTer Stranded Total RNA-Seq Kit v2 (Clontech Laboratories, USA) with slight modifications. Post-ligated material was individually barcoded with unique dual-indexed primers. The concentration of the libraries was then measured by PicoGreen assay (Thermo Fisher Scientific, USA), and the average fragment sizes of the libraries were estimated by

using a DNA High Sense chip on a LabChip GX Touch Nucleic Acid Analyzer (PerkinElmer, USA). KAPA quantitative PCR assay (Roche, USA) was performed to assess the nanomolar amounts of ligated libraries. Final libraries were pooled equimolar and underwent 2x100 bp paired-end sequencing on a NovaSeq 6000 instrument using an S4 200-cycle flow cell to achieve the desired read depth. RNA sequencing was performed at Discovery Life Sciences (USA).

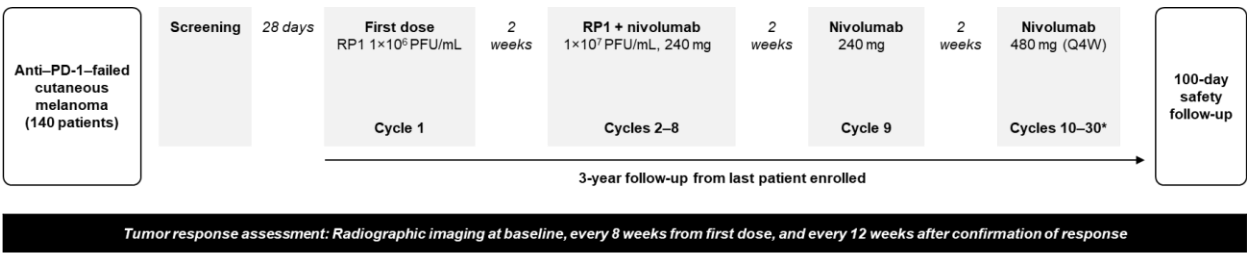
Transcript-level quantification was performed using salmon quant¹ v1.10.1 against GRCh38 (Ensembl release 103). Generated abundance estimates were aggregated using the tximport v1.34 R package,² and differential expression analysis was performed using DESeq2 with prior filtering for pseudogenes and low expression³ (cutoff for downstream analysis was >10 reads in at least five samples). Over-representation analysis (Gene Ontology Biological Processes [GO:BP]) was performed on significant differentially expressed genes (DEGs; **P**-adj <0.05 [Benjamini-Hochberg]) using the genekitr v1.2.8 and geneset v0.2.7 packages,⁴ plotting the top 30 significantly enriched terms by FoldEnrich, GeneRatio, and RichFactor above their median values. The most upregulated DEGs (**P**-adj <0.05, log₂FC >1.5, excluding T-cell receptor and B-cell receptor genes) between the pre-treatment and on-treatment samples in responding patients (partial response + complete response) were incorporated into a treatment-induced response signature and associated with response within RNA sequencing data using the SingScore gene signature scoring method.⁵

Injected vs Non-Injected Lesion Analysis

An electronic data capture (EDC) database was used to assess lesions injected per patient, including size and anatomic description and location. Image snapshots from blinded independent central reviewers were then compared with the EDC listing. When injected lesion listings from the EDC were unclear, direct communication with the clinical trial sites was made for clarification.

SUPPLEMENTAL FIGURES AND TABLES

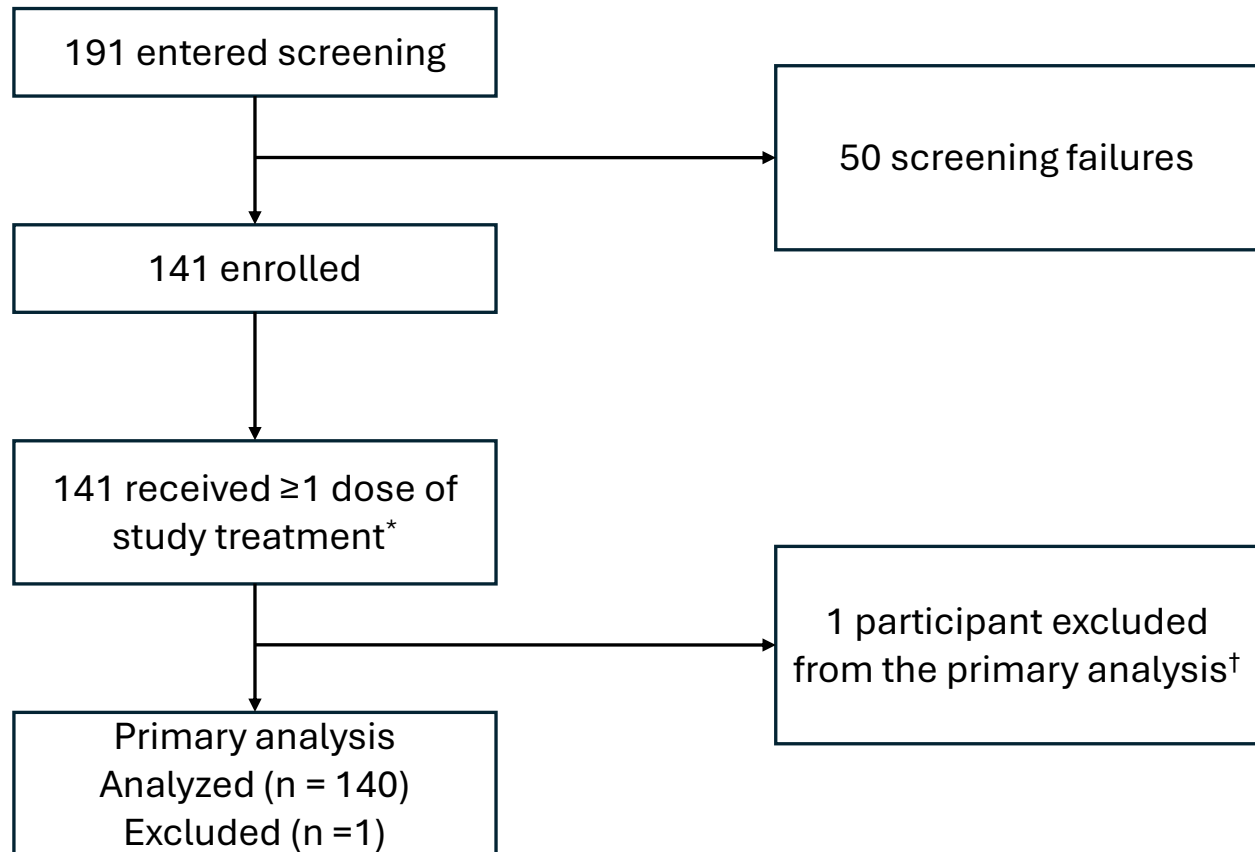
FIG S1. IGNYTE study design.



Primary objective	Safety and efficacy using mRECIST [†] 1.1 by independent central review (also assessed by RECIST 1.1)
Secondary objective	• ORR by investigator assessment (mRECIST[†] 1.1) • DOR, CR rate, and PFS by central and investigator assessment; 1-year and 2-year OS
Key eligibility	Anti-PD-1-failed advanced melanoma; measurable disease; adequate organ function; no prior oncolytic therapy; ECOG performance status 0–1
Criteria for prior anti-PD-1-failure	Confirmed progression while being treated with at least 8 weeks of anti-PD-1 therapy, alone or in combination; anti-PD-1 must be the last prior therapy. Patients on prior adjuvant therapy must have confirmed progression <u>while being treated with</u> adjuvant treatment (PD can be confirmed by biopsy)

*Further RP1 can be administered beyond eight cycles if protocol-specified criteria are met. [†]For mRECIST, PD must be confirmed by further progression at least 4 weeks after initial PD to better allow for pseudoprogression compared with RECIST 1.1. CR, complete response; DOR, duration of response; ECOG, Eastern Cooperative Oncology Group; mRECIST, modified RECIST 1.1; ORR, objective response rate; OS, overall survival; PD, progressive disease; PD-1, programmed cell death protein 1; PFS, progression-free survival; PFU, plaque-forming units; Q4W, every 4 weeks; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1.

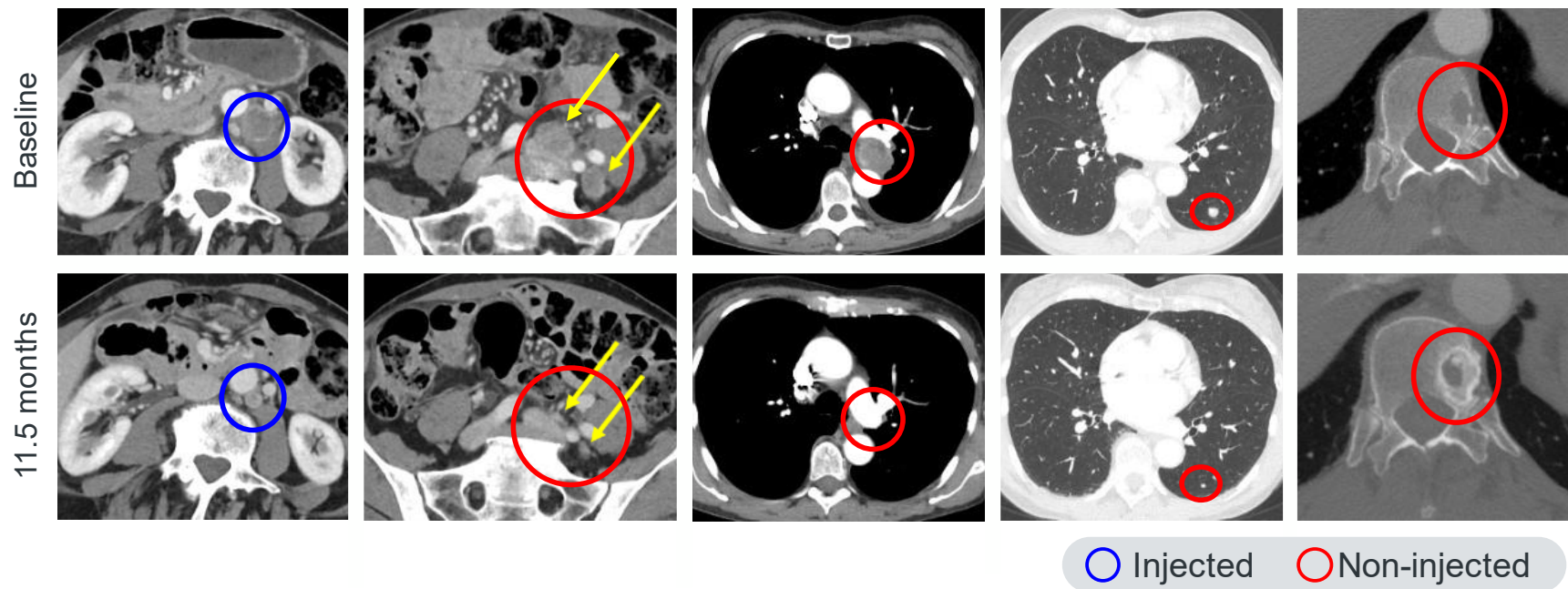
FIG S2. CONSORT diagram.



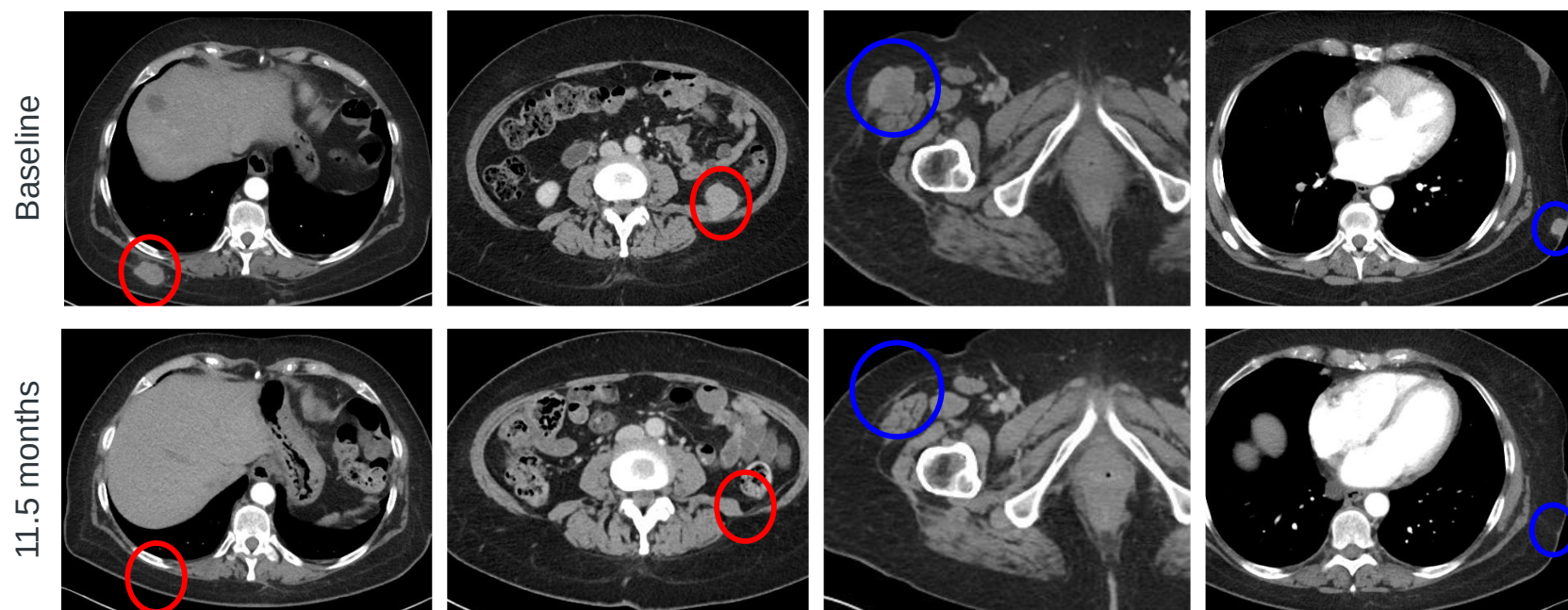
*All 141 patients in this cohort received at least 1 dose of RP1, and 138 patients received at least 1 dose of nivolumab. †One patient with a history of melanoma was found to have active sarcoidosis (non-cancerous condition) by biopsy at cycle 1 day 1.

FIG S3. Patient examples of the outcome of RP1 plus nivolumab.

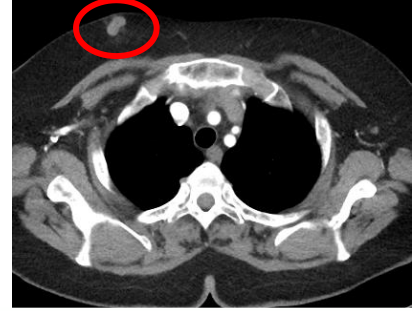
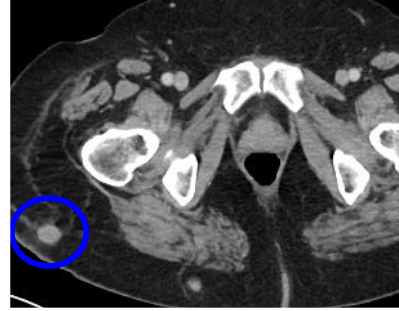
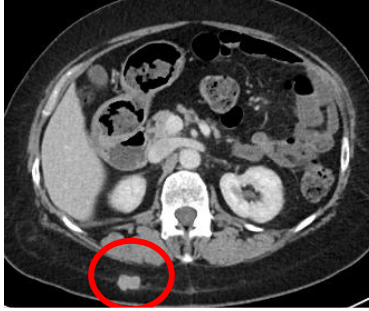
A) Patient example 1 (stage IVM1b).



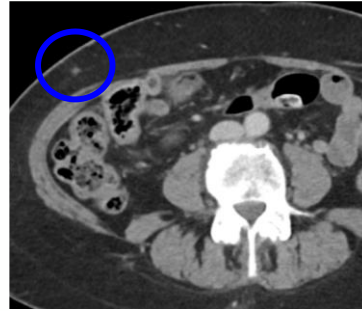
B) Patient example 2 (stage IVM1a).

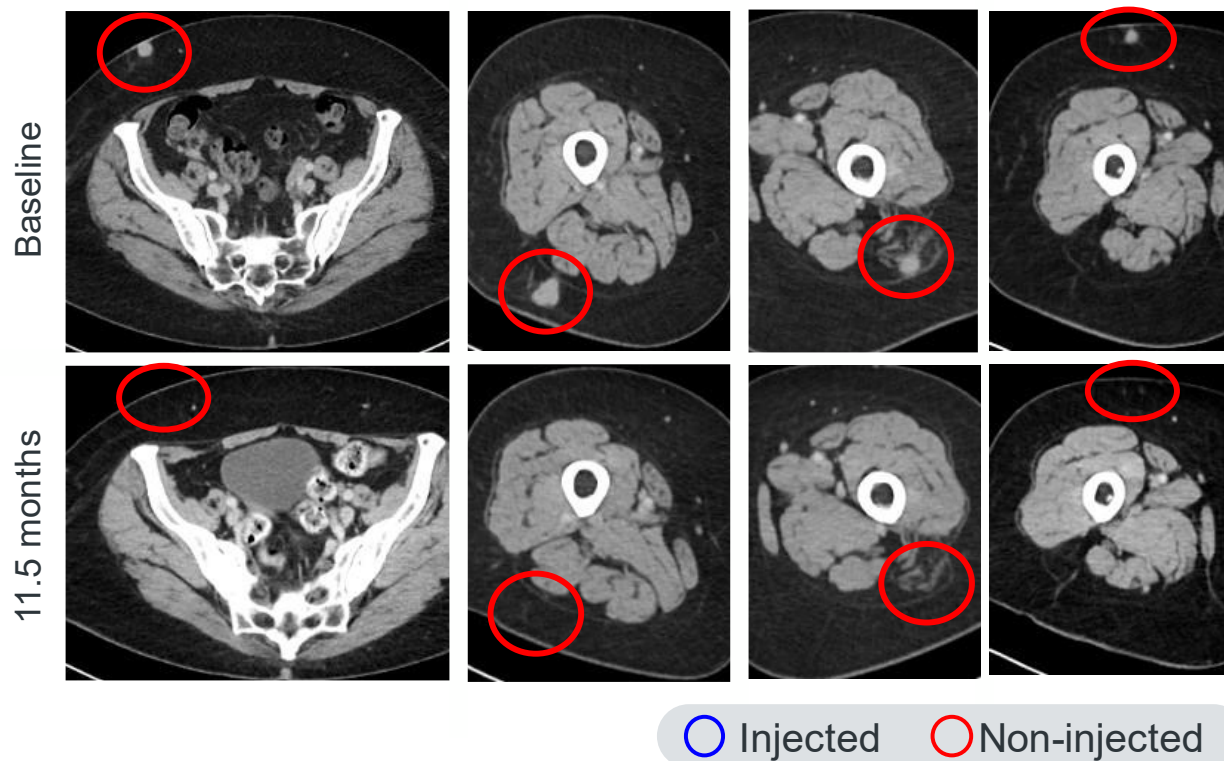


Baseline



11.5 months





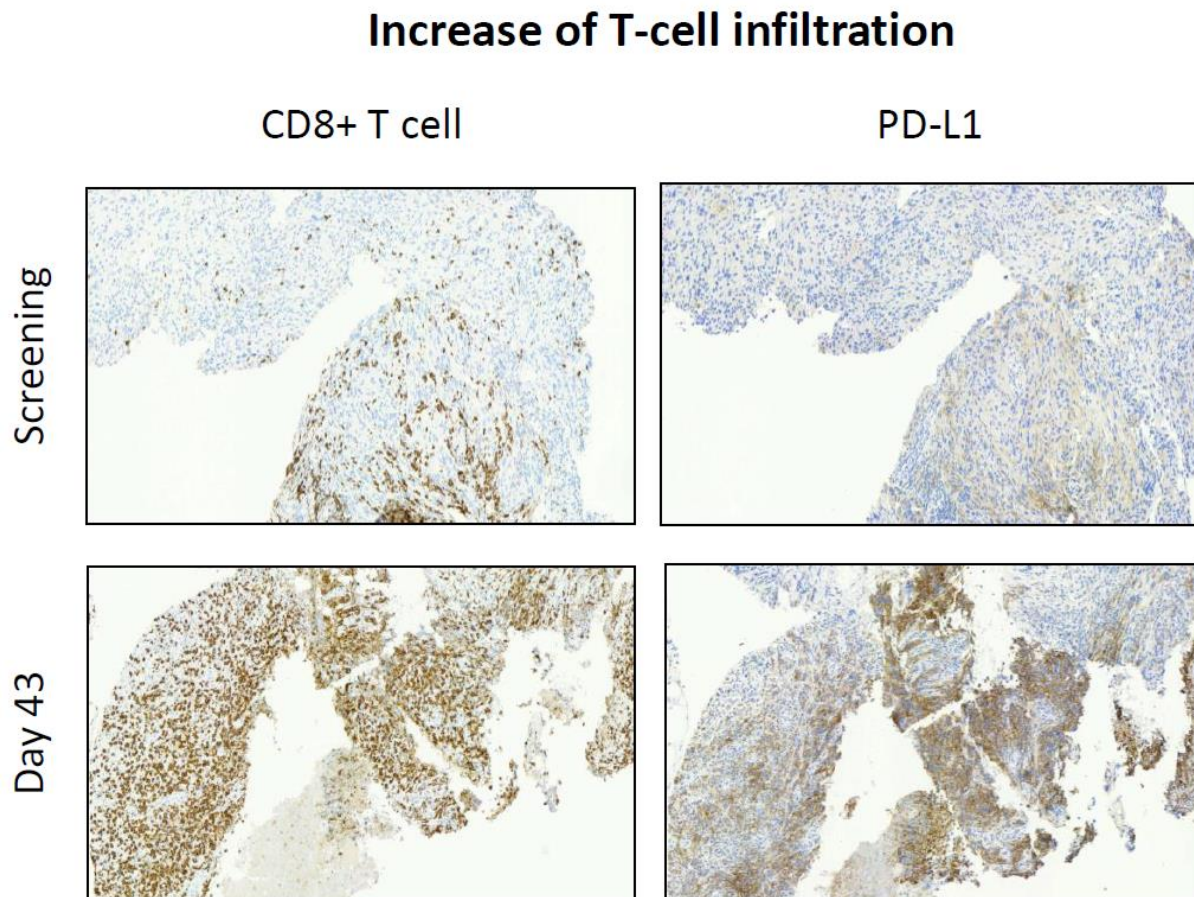
Panel A shows a patient with stage IVM1b disease (yellow arrows in panel A denote retroperitoneal lymph nodes) and panel B shows a patient with stage IVM1a disease. Both patients experienced responses in multiple non-injected distant tumors.

FIG S4. RP1 combined with nivolumab increases CD8+ T-cell infiltration and PD-L1 expression.

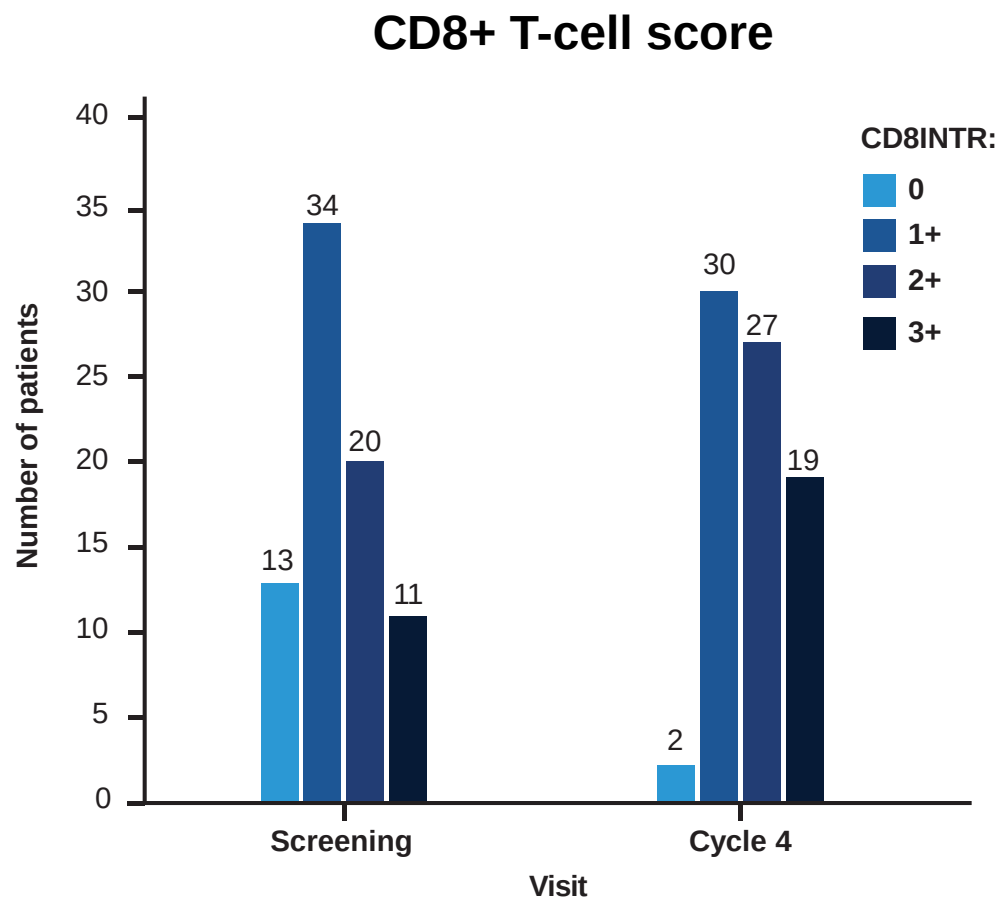
A) Immunohistochemistry of CD8 and PD-L1 at screening and day 43 of treatment.

Quantification of B) CD8+ T-cell score and C) PD-L1 CPS.

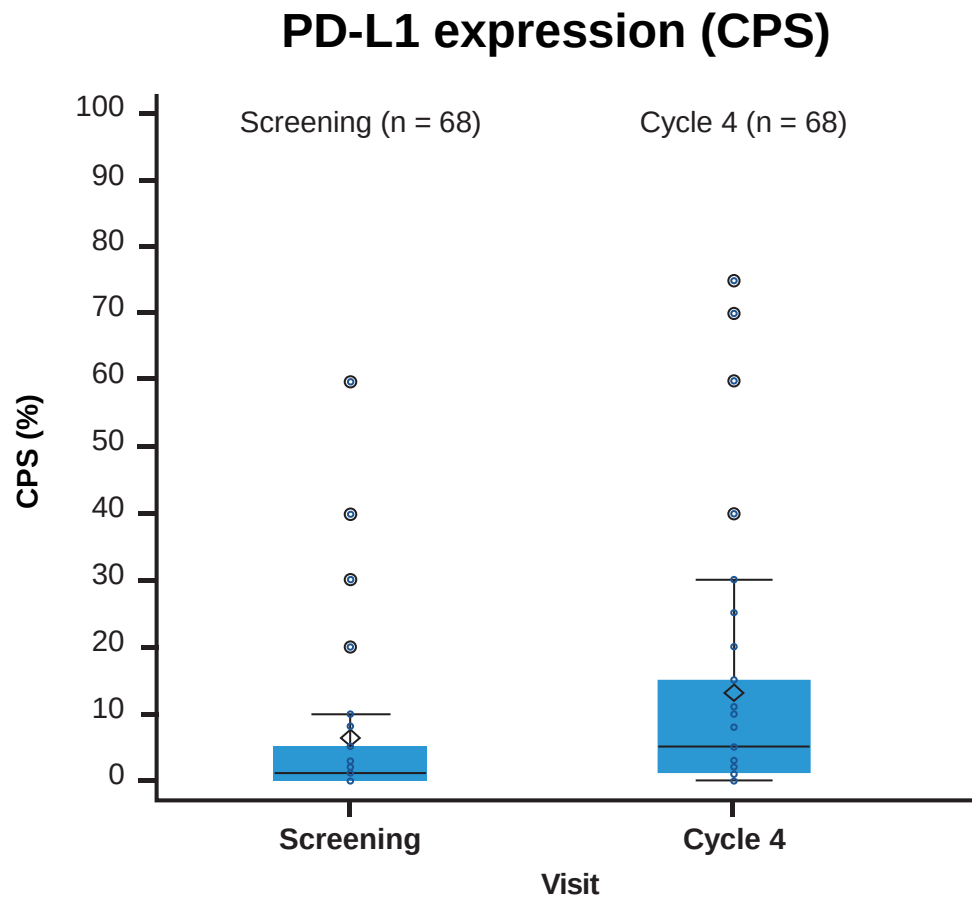
A)



B)



C)



CD, cluster of differentiation; CD8INTR, intratumoral CD8 score; CPS, combined positive score; PD-L1, programmed death-ligand 1.

TABLE S1. Summary of Dose Volume in the Initial Course.

RP1 Injection Volume, mL	
N	140
Mean (Std)	22.9 (20.36)
Median	17.4
Min, max	1.0, 80.0

Abbreviations: max, maximum; min, minimum; Std, standard deviation.

TABLE S2. Summary of Best Reduction in Injected and Non-Injected Lesions Among Responding Patients by RECIST 1.1.

	Number (%) of Measured Lesions for Responders by RECIST 1.1 (CR or PR; N = 46)	
	Injected Lesions (n = 78)	Non-Injected Lesions (n = 119)
No. (%) of lesions with		
No reduction	1 (1.3)	4 (3.4)
Any reduction	77 (98.7)	115 (96.6)
Best reduction ≥ 0 to $<30\%$	4 (5.1)	21 (17.6)
Best reduction ≥ 30 to $<100\%$	31 (39.7)	47 (39.5)
Best reduction of 100%	42 (53.8)	47 (39.5)

Abbreviations: CR, complete response; PR, partial response; RECIST 1.1, Response

Evaluation Criteria in Solid Tumors version 1.1.

TABLE S3. Summary of Tumor Reduction in Non-Injected Lesions in Visceral Organs of Responding Patients by RECIST 1.1.

Location of Visceral Lesion	Lesions, n	Any Reduction, No. (%)	<30%, No. (%)	≥30% to <100%, No. (%)	100%, No. (%)
Lung	29	28 (96.6)	7 (24.1)	9 (31.0)	12 (41.4)
Liver	14	14 (100.0)	3 (21.4)	5 (35.7)	6 (42.9)
Spleen	6	5 (83.3)	5 (83.3)	0	0
Pleura	2	2 (100.0)	0	1 (50.0)	1 (50.0)
Brain	1	1 (100.0)	1 (100.0)	0	0
Total	52	50 (96.2)	16 (30.8)	15 (28.8)	19 (36.5)

Abbreviations: RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1.

TABLE S4. Summary of Responses by Injection Type by RECIST 1.1.

Confirmed BOR, No. (%)	Superficial Only (n = 104)	Visceral and Deep (n = 22)	Visceral and Deep Plus Superficial (n = 14)
CR	18 (17.3)	3 (13.6)	0
PR	13 (12.5)	6 (27.3)	6 (42.9)
SD	19 (18.3)	8 (36.4)	4 (28.6)
PD	46 (44.2)	5 (22.7)	3 (21.4)
NE	8 (7.7)	0	1 (7.1)
ORR, No. (%)	31 (29.8)	9 (40.9)	6 (42.9)

Abbreviations: BOR, best overall response; CR, complete response; NE, not evaluable;

ORR, objective response rate; PD, progressive disease; PR, partial response; RECIST

1.1, Response Evaluation Criteria in Solid Tumors version 1.1; SD, stable disease.

TABLE S5. BOR Among All Patients and Across Subgroups by Blinded Independent Central Review According to mRECIST 1.1.

BOR No. (%)	All Patients (N = 140)	Prior Anti-PD-1		Stage		Anti-PD-1 Resistance		Anti-PD-1 Adjuvant		BRAF		PD-L1 Expression ^a	
		Without Anti-CTLA-4 ^b (n = 75)	With Anti-CTLA-4 (n = 65)	IIIB/IIIC/IVM1a (n = 72)	IVM1 b/c/d (n = 68)	Primary (n = 92)	Secondary (n = 48) ^c	Yes (n = 36)	No (n = 104)	Mut (n = 53)	WT (n = 87)	Pos. (n = 44)	Neg. (n = 79)
CR	21 (15.0)	16 (21.3)	5 (7.7)	17 (23.6)	4 (5.9)	16 (17.4)	5 (10.4)	11 (30.6)	10 (9.6)	8 (15.1)	13 (14.9)	11 (25.0)	10 (12.7)
PR	26 (18.6)	13 (17.3)	13 (20.0)	12 (16.7)	14 (20.6)	17 (18.5)	9 (18.8)	5 (13.9)	21 (20.2)	11 (20.8)	15 (17.2)	12 (27.3)	10 (12.7)
SD	41 (29.3)	20 (26.7)	21 (32.3)	24 (33.3)	17 (25.0)	22 (23.9)	19 (39.6)	10 (27.8)	31 (29.8)	19 (35.8)	22 (25.3)	8 (18.2)	27 (34.2)
PD	43 (30.7)	24 (32.0)	19 (29.2)	18 (25.0)	25 (36.8)	31 (33.7)	12 (25.0)	9 (25.0)	34 (32.7)	12 (22.6)	31 (35.6)	10 (22.7)	27 (34.2)
NE	9 (6.4)	2 (2.7)	7 (10.8)	1 (1.4)	8 (11.8)	6 (6.5)	3 (6.3)	1 (2.8)	8 (7.7)	3 (5.7)	6 (6.9)	3 (6.8)	5 (6.3)
ORR (95% CI)	33.6 (25.8, 42.0)	38.7 (27.6, 50.6)	27.7 (17.3, 40.2)	40.3 (28.9, 52.5)	26.5 (16.5, 38.6)	35.9 (26.1, 46.5)	29.2 (17.0, 44.1)	44.4 (27.9, 61.9)	29.8 (21.2, 39.6)	35.8 (23.1, 50.2)	32.2 (22.6, 43.1)	52.3 (36.7, 67.5)	25.3 (16.2, 36.4)

Abbreviations: BOR, best overall response; CI, confidence interval; CR, complete response; CTLA-4, cytotoxic T-

lymphocyte antigen 4; mRECIST 1.1, modified Response Evaluation Criteria in Solid Tumors version 1.1; Mut, mutant;

NE, not evaluable; Neg, negative; ORR, objective response rate; PD, progressive disease; PD-1, programmed cell death protein 1; PD-L1, programmed death-ligand 1; Pos, positive; PR, partial response; SD, stable disease; WT, wild-type.

^aThere were 17 patients with unknown PD-L1 expression status.

^bIncludes 66 patients receiving anti-PD-1 as monotherapy (ORR 40.9%, CR rate 22.7%). Nine patients received anti-PD-1 in combination with other therapeutic agents.

^cIncludes one patient with unknown resistance status.

TABLE S6. Full List of Grade 3/4 TRAEs.

	All Patients (N = 140)
Any grade 3 or 4 TRAE	18 (12.9)
TRAEs by preferred term	
Abdominal pain	1 (0.7)
Acute left ventricular failure	1 (0.7)
Amylase increased	1 (0.7)
Arthralgia	1 (0.7)
Asthenia	1 (0.7)
Cancer pain	1 (0.7)
Cytokine release syndrome	1 (0.7)
Decreased appetite	1 (0.7)
Diarrhea	1 (0.7)
Eczema	1 (0.7)
Enterocolitis	1 (0.7)
Extranodal marginal zone B-cell lymphoma (MALT type)	1 (0.7)
Fatigue	1 (0.7)
Hepatic cytolysis	1 (0.7)
Hyponatremia	1 (0.7)
Hypophysitis	2 (1.4)
Immune-mediated enterocolitis	1 (0.7)
Infusion-related reaction	1 (0.7)
Left ventricular dysfunction	1 (0.7)
Lipase increased	1 (0.7)
Memory impairment	1 (0.7)
Meningitis aseptic	1 (0.7)
Muscular weakness	1 (0.7)
Myocarditis	1 (0.7)
Palmar-plantar erythrodysesthesia syndrome	1 (0.7)
Paresthesia	1 (0.7)
Peripheral sensory neuropathy	1 (0.7)
Radiculitis brachial	1 (0.7)
Rash maculo-papular	2 (1.4)
Sinus arrhythmia	1 (0.7)
Splenic rupture	1 (0.7)
Tricuspid valve incompetence	1 (0.7)
Tumor pain	1 (0.7)
Type 1 diabetes mellitus	1 (0.7)

Data shown as No. (%). In total, there were five grade 4 TRAEs: cytokine release

syndrome, hepatic cytolysis, lipase increased, myocarditis, and splenic rupture (n = 1 each). TRAEs were graded according to CTCAE version 5.0.

Abbreviations: CTCAE, Common Terminology Criteria for Adverse Events; MALT, mucosa-associated lymphoid tissue; TRAE, treatment-related adverse event.

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